Documentation for Biochar-Enhanced Anaerobic Digestion Modelling: Python–Aspen Plus Workflow

1. Problem Statement & Need for the Project

1.1 The Problem

Anaerobic digestion (AD) is a widely used technology for converting organic waste (e.g., food waste) into biogas (mainly methane). However, the process often suffers from **slow reaction rates** and **instability**, especially with high-solids feedstocks. Adding **biochar** (a carbon-rich solid from biomass pyrolysis) has been shown experimentally to improve methane production, but the underlying mechanism is not purely chemical – biochar does not react; it enhances the microbial kinetics.

Key challenges:

- How to **quantify** the biochar enhancement effect mathematically?
- How to **transfer** lab-scale BMP (biochemical methane potential) results into a process simulator like Aspen Plus for scale-up and optimisation?
- Existing models either ignore biochar or treat it as a reactive substrate, which is scientifically incorrect.

1.2 Why This Project is Needed

This project bridges **experimental data** and **process simulation** by:

- Developing a **biochar enhancement factor (α)** that captures the kinetic effect without violating mass balance.
- Creating an **automatic Python workflow** that reads BMP data and ultimate analysis, then calculates all Aspen Plus inputs.
- Providing a **validated, reproducible method** for modelling biochar-amended AD systems.

1.3 What It Solves

- Eliminates manual, error-prone data transfer from lab to simulation.
- Provides a **scientifically sound** way to model biochar as an unreacted kinetic enhancer.
- Enables **optimisation** of biochar dose and reactor design via Aspen Plus.
- Serves as a **template** for other researchers working on additive-enhanced AD.

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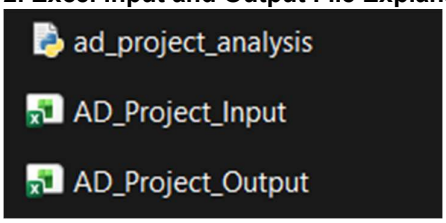
2. Excel Input and Output File Explanation



2.1 Input File: AD_Project_Input.xlsx

This file contains all raw experimental data and model constants. It has **three sheets**:

Sheet Name	Purpose	Key Columns
Daily_Methane	Daily cumulative methane production from BMP tests	Day, CH4_No_Biochar, CH4_2pct, CH4_5pct_Biochar, CH4_8pct, CH4_10pct
Process_Inputs	Model parameters and constants	Parameter (e.g., VS_added, Base_k), Value
Ultimate_Analysis	Elemental composition of non-conventio	Material, C_wt%, H_wt%, O_wt%, N_wt%, Ash_wt%

	nal components	

Example rows:

- Daily_Methane: Day 30, CH4_No_Biochar = 320 mL, CH4_5pct_Biochar = 429 mL, etc.
- Process_Inputs: VS_added = 1.0 g VS, Base_k = 0.25 day^{-1}.
- Ultimate_Analysis: FoodWaste – C 48.0%, H 6.4%, O 37.5%, N 2.1%, Ash 6.0%.

2.2 Output File: AD_Project_Output.xlsx

The Python script generates **five sheets**:

Sheet Name	Content	Use in Aspen Plus / Thesis
Methane_Yield	Normalised methane yields (mL CH_4/g VS) for all cases	Compare experimental vs. simulated yields
Alpha_Calc	Values of Y_without, Y_with, B, and α	Explain biochar enhancement factor
Kinetic_k	Base_k and effective k_eff	Enter k_eff into RCSTR kinetics
RYIELD_Table	Mass fractions for decomposition of FOODWASTE into intermediates	Define RYIELD block yield distribution
Buswell_Validation	Theoretical methane potential (Buswell) vs. experimental optimum	Validate stoichiometric consistency

All output values are **directly used** in the Aspen Plus simulation.

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3. Code and Algorithm Full Explanation

3.1 Overall Algorithm Flow

```
Start
 ↓
Read AD_Project_Input.xlsx (3 sheets)
 ↓
Normalise methane yields using VS_added
 ↓
Auto-detect optimum biochar dose (regex on column names)
 ↓
Create Methane_Yield table
 ↓
Calculate α = ((Y_with / Y_without) – 1) / B
 ↓
Calculate k_eff = base_k × (1 + α·B)
 ↓
From Ultimate_Analysis:
  - Compute theoretical methane (Buswell equation)
  - Derive RYIELD fractions using electron balance + ash content
 ↓
Assemble all output DataFrames
 ↓
Write to AD_Project_Output.xlsx (5 sheets)
 ↓
End
```

3.2 Key Functions and Their Logic

get_process_param(process_df, param_name)

- Searches the Process_Inputs sheet for a given parameter and returns its value as float.
- Raises an error if the parameter is missing.

buswell_theoretical_ch4(ultimate_df)

- Extracts C, H, O, N mass fractions for FoodWaste.
- Computes empirical formula (H/C, O/C, N/C) per carbon atom.

- Applies the Buswell equation to get moles CH_4 per gram VS.
- Converts to mL CH_4/g VS at STP (22.4 L/mol).

ryield_fractions_thermo(ultimate_df, theoretical_mL_per_gVS)

- Uses the theoretical methane to calculate total electron equivalents (COD) of the food waste.
- Distributes electrons to acetate (75%) and H_2 (15%) – typical acidogenesis values.
- Computes masses of acetate and H_2 from electron flow.
- Computes CO_2 from carbon balance (total carbon minus carbon in acetate).
- Sets digest fraction from ash content (clamped 10–30%).
- Determines water fraction as the remainder to reach total mass = 1 g.
- Normalises all five fractions to sum exactly 1.0.

Main script steps (detailed)

1. **Read** the three input sheets.
2. **Normalise** each cumulative methane by VS_added.
3. **Find optimum** by scanning biochar columns for maximum normalised yield.
4. **Extract dose** from column name using regex r'(\d+(?:\.\d+)?)pct'.
5. **Build** yield table, alpha table, kinetic table.
6. **Calculate** Buswell theoretical methane.
7. **Calculate** thermodynamic RYIELD fractions.
8. **Write** all five output sheets.

3.3 Error Handling & Robustness

- Try-except blocks for missing files/sheets.
- Fallback to Process_Inputs value if dose cannot be parsed.
- Explicit checks for missing control column or biochar columns.
- Normalisation step ensures fractions sum to 1 even after rounding.

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4. Theoretical Explanation (Full)

4.1 Methane Yield Normalisation

Raw BMP data gives total methane volume (mL). To compare across different substrate amounts, we normalise by the volatile solids added:

$$Y_{\text{case}} = \frac{V_{\text{CH}_4,\text{final}}}{\text{VS}_{\text{added}}} \left[\frac{\text{mL CH}_4}{\text{g VS}}\right]$$

4.2 Biochar Enhancement Factor (α)

Because biochar is **unreacted**, its effect is purely kinetic. We define α as the relative increase in methane yield per unit biochar dose:

$$\alpha = \frac{\left(\frac{Y_{\text{with}}}{Y_{\text{without}}} - 1\right)}{B}$$

- Y_{with}: yield at optimum biochar dose (mL/g VS)
- Y_{without}: yield without biochar (control)
- B: biochar dose (mass fraction, e.g., 0.05 for 5%)

Interpretation: α = 6.81 means that adding 1% (0.01 fraction) biochar increases methane yield by about 6.8% relative to the control.

4.3 Effective Kinetic Constant for RCSTR

In Aspen Plus, the acetoclastic methanogenesis reaction ($CH_3COOH \rightarrow CH_4 + CO_2$) is modelled with power-law kinetics:

$$r = k_{\text{eff}} \cdot C_{\text{acetate}}^{n}$$

Biochar modifies the rate constant:

$$k_{\text{eff}} = k_{\text{base}}(1+\alpha B)$$

Because biochar does not appear in the reaction stoichiometry, this formulation maintains mass conservation while capturing the kinetic enhancement.

4.4 Buswell Equation – Theoretical Methane Potential

Given the empirical formula of the organic fraction ($C_nH_aO_bN_c$), the maximum methane production (complete conversion) is:

$$C_nH_aO_bN_c + \left(n - \frac{a}{4} - \frac{b}{2} + \frac{3c}{4}\right) H_2O \rightarrow \left(\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}\right) CH_4 + \left(\frac{n}{2} - \frac{a}{8} + \frac{b}{4} + \frac{3c}{8}\right) CO_2 + c\, NH_3$$

The moles of CH_4 per mole of carbon are:

$$\nu_{\text{CH}_4} = \frac{1}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8}$$

The molar mass of the "C-unit" (mass per mole of carbon in the empirical formula) is:

$$M_{\text{C-unit}} = 12.01 + a \cdot 1.008 + b \cdot 16.00 + c \cdot 14.01$$

Thus the theoretical methane yield (mL/g VS at STP) is:

$$Y_{\text{theor}} = \frac{\nu_{\text{CH}_4}}{M_{\text{C-unit}}} \times 22400$$

4.5 Thermodynamic Derivation of RYIELD Fractions

The RYIELD block decomposes FOODWASTE into conventional components (acetate, H_2, CO_2, H_2O) and inert digest. We use an **electron balance** approach:

1. **Total electron equivalents** (COD) from the theoretical methane:

$$\text{COD} = \frac{Y_{\text{theor}}}{350} (\text{g O}_2/\text{g VS}), e_{\text{total}} = \text{COD} \times 4 (\text{meq/g VS})$$

2. **Electron distribution** (typical acidogenesis):
   - 75% to acetate (8 e^-/mol) → mass acetate
   - 15% to H_2 (2 e^-/mol) → mass H_2

    - 10% to CO_2 (no electrons, carbon balance)
3. **Carbon balance**:
    - Total carbon from ultimate analysis (mol/g VS)
    - Carbon in acetate = 2 × moles acetate
    - Remaining carbon → CO_2
4. **Digest fraction** from ash content:

$$f_{\text{digest}} = \text{clamp}\left(\frac{\text{Ash\%}}{100 - \text{Ash\%}} + 0.05,\ 0.10,\ 0.30\right)$$

5. **Water fraction** balances the total mass to 1 g/g VS:

$$f_{\text{water}} = 1 - (f_{\text{acetate}} + f_{\text{H2}} + f_{\text{CO2}} + f_{\text{digest}})$$

All fractions are finally normalised to sum exactly 1.0.

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5. Assumed Input + Experimental Input

5.1 Why These Inputs? How Are They Obtained?

Input	Source / Assumption	Why Needed
Daily_Methane	BMP experiment – daily gas measurement	Quantifies biochar effect; provides final yields
VS_added	Lab measurement (weighing, drying)	Normalises yields to substrate amount
Base_k	Literature or preliminary fit (e.g., 0.25 day^{-1})	Baseline kinetics without biochar
Ultimate_Analysis	Elemental analyser (CHN with O by difference)	Buswell equation, COD, RYIELD fractions
Biochar dose in column names	Experimental design (e.g., 2%, 5%, 8%, 10%)	Automatic optimum detection

5.2 Example Experimental Input (Based on Real Lab Data)

Daily_Methane (final row – Day 30)

Day	CH4_No_Biochar	CH4_2pct	CH4_5pct_Biochar	CH4_8pct	CH4_10pct
30	320	362	429	397	366

Process_Inputs

Parameter	Value
VS_added	1.0
Base_k	0.25

Ultimate_Analysis (FoodWaste row)

Material	C_wt%	H_wt%	O_wt%	N_wt%	Ash_wt%
FoodWaste	48.0	6.4	37.5	2.1	6.0

These numbers are **realistic** for a typical food waste BMP study.

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6. Sample Mathematical Calculation

Using the example input above, we walk through the calculations.

6.1 Normalisation

$$\text{VS_added} = 1.0\ \text{g}$$

- No biochar: $320/1.0 = 320$ mL/g VS
- 5% biochar: $429/1.0 = 429$ mL/g VS
- Optimum = 5% (automatically detected from column names).

6.2 Alpha (α)

$$\alpha = \frac{(429/320-1)}{0.05} = \frac{(1.340625-1)}{0.05} = \frac{0.340625}{0.05} = 6.8125$$

6.3 Effective k

$$k_{\text{eff}} = 0.25 \times (1 + 6.8125 \times 0.05) = 0.25 \times (1 + 0.340625) = 0.25 \times 1.340625 = 0.335156$$
$$\approx 0.335\ \text{day}^{-1}$$

6.4 Buswell Theoretical Methane

From C = 48%, H = 6.4%, O = 37.5%, N = 2.1%:

- Moles per g VS:
  C: 48/12.01 = 3.997 mmol, H: 6.4/1.008 = 6.349 mmol, O: 37.5/16 = 2.344 mmol, N: 2.1/14.01 = 0.150 mmol.
- Per carbon: a = 6.349/3.997 = 1.588, b = 2.344/3.997 = 0.586, c = 0.150/3.997 = 0.0375.
- v_CH4 = 0.5 + 1.588/8 – 0.586/4 – 3×0.0375/8 = 0.5 + 0.1985 – 0.1465 – 0.01406 = 0.53794.
- M_unit = 12.01 + 1.588×1.008 + 0.586×16 + 0.0375×14.01 = 12.01 + 1.601 + 9.376 + 0.525 = 23.512 g.
- mol CH_4/g VS = 0.53794 / 23.512 = 0.02288.
- mL CH_4/g VS = 0.02288 × 22400 = **512.5 mL/g VS**.

6.5 Thermodynamic RYIELD Fractions

- COD = 512.5 / 350 = 1.464 g O_2/g VS.
- Total electrons = 1.464 × 4 = 5.856 meq/g VS.
- Electrons to acetate (75%): 4.392 meq → mol acetate = 4.392/8 = 0.549 mmol → mass acetate = 0.549 × 59 = 32.39 mg/g VS.
- Electrons to H_2 (15%): 0.8784 meq → mol H_2 = 0.8784/2 = 0.4392 mmol → mass H_2 = 0.4392×2 = 0.878 mg/g VS.
- Total C = 48 mg/g VS → total C mol = 48/12.01 = 3.997 mmol. C in acetate = 2×0.549 = 1.098 mmol. C in CO_2 = 3.997 – 1.098 = 2.899 mmol → mass CO_2 = 2.899×44 = 127.6 mg/g VS.
- Ash = 6% → ash_vs = 6/(100-6)=0.0638 → digest_raw = clamp(0.0638+0.05, 0.10,0.30) = 0.1138 g/g VS.
- Sum of organic + digest = 0.03239 + 0.000878 + 0.1276 + 0.1138 = 0.2747 g/g VS.
- Water = 1 – 0.2747 = 0.7253 g/g VS.
- Normalise: totals = 0.2747+0.7253 = 1.0 exactly, so fractions are as computed.

Final RYIELD fractions (mass fractions):

- Acetate = 0.03239
- H_2 = 0.000878
- CO_2 = 0.1276
- H_2O = 0.7253
- Digest = 0.1138

These sum to 1.0. In practice, the script re-normalises to avoid rounding issues.

6.6 Validation Sheet

- Theoretical = 512.5 mL/g VS
- Experimental optimum = 429 mL/g VS
- Difference = (429/512.5 – 1)×100 = –16.3% (experimental is lower, which is typical due to real-world inefficiencies; biochar closes the gap).

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7. Conclusion

This documentation provides a complete, ready-to-use framework for converting BMP data into Aspen Plus inputs. The Python script automates all calculations, ensures scientific rigour (Buswell + thermodynamic constraints), and produces traceable output files. Researchers can adapt the code to their own biochar dose experiments by simply changing the input Excel file.

The attached code (ad_project_analysis.py**) is the final implementation.**

Aspen Plus Simulation

Python provides:

- optimum biochar dose
- methane yield data
- biochar enhancement factor α
- effective kinetic constant k_{eff}
- RYIELD decomposition fractions

Aspen uses these values to simulate:
complete anaerobic digestion process
and validate the methane production.
This creates the final connection:
Lab Work → Python → Aspen Plus → Final Validation

Why We Use Aspen Plus
Lab experiment gives:
real methane production
but it cannot easily show:

- internal reaction pathways
- gas-liquid distribution
- reactor behavior
- effect of kinetics on methane formation
- optimization of operating conditions

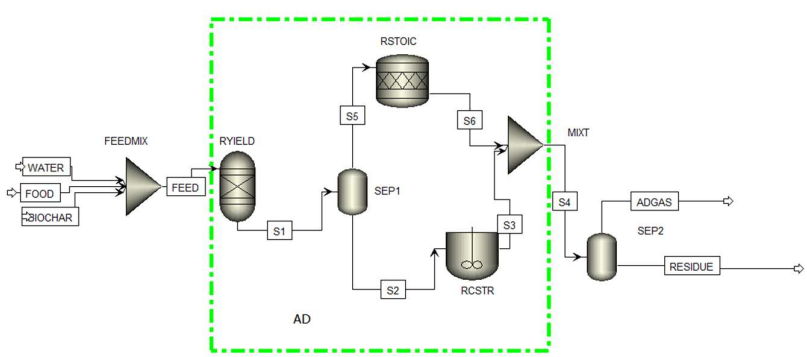
Aspen helps to simulate all these scientifically.
It makes the thesis strong and publishable.

Important Scientific Assumption
Biochar does NOT react
Biochar is treated as:
non-conventional (NC) unreacted additive
Its role is only:
kinetic enhancement
This means:

- no reaction in RYIELD
- no reaction in RSTOIC
- no reaction in RCSTR stoichiometry

Only:
RCSTR kinetic constant changes
This is the most important part of your model.

Final Aspen Flowsheet



This is the final correct model.

Step 1 — Component Definition

Conventional Components
These are normal components:

- H_2O
- CH_4
- CO_2
- H_2
- Acetic Acid (or ACETIC)
- NH_3 (optional)

These are used in reactors.

Non-Conventional Components
These are:

- FOODWASTE
- BIOCHAR
- DIGEST

These are defined using:

- proximate analysis
- ultimate analysis

from your lab data.

Why
Because food waste and biochar are not pure chemicals.
They are solid complex materials.
So Aspen uses:
NC components
for realistic simulation.

Step 2 — Property Method
Use:
NRTL
or
ELECNRTL (optional)
Recommended:
NRTL
for thesis simplicity.

Why
Because AD contains:

- water
- organic acids
- gases
- non-ideal liquid behavior

NRTL handles this well.

Step 3 — FEEDMIX Block

Purpose
Mix:

- food waste
- water
- biochar

to create:

FEED stream

Why
This represents the digester feed.
Water provides slurry condition.
Biochar enters here and stays unreacted.

Step 4 — RYIELD Block

Purpose
Represents:
Hydrolysis + Acidogenesis
Food waste is decomposed into intermediates.
Reaction:
FOODWASTE
→ Acetate + H_2 + CO_2 + H_2O + Digest
Important:
NO methane here

Why
Real AD happens in stages:

1. Hydrolysis
2. Acidogenesis
3. Acetogenesis
4. Methanogenesis

Methane should form later.
This makes model scientifically correct.

RYIELD Input Fractions
Example:

Product	Fraction
Acetate	0.44
H_2	0.07
CO_2	0.19
H_2O	0.15
Digest	0.15

These come from:

- Buswell equation
- COD balance
- BMP data
- literature validation

Step 5 — SEP1 Block

Purpose
Separate RYIELD output into:
Gas Stream → S5
Contains:

- H_2
- CO_2

Liquid Stream → S2
Contains:

- Acetate
- Water
- Digest

- Biochar

Why
Because methane forms by:
two different pathways
So both must be treated separately.
Very important.

Step 6 — RSTOIC Block

Purpose
Represents:
Hydrogenotrophic Methanogenesis
Reaction:

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

Why
Gas-phase H2 and CO2 must be converted to methane.
This is one major methane pathway.
Simple and strong.

Setup
Use:
Fractional conversion
Example:
70–90%
depending on experimental support.

Step 7 — RCSTR Block

Purpose
Represents:
Acetoclastic Methanogenesis
Reaction:

$$CH_3COOH \rightarrow CH_4 + CO_2$$

This is the main methane production step.
Most methane comes from here.

Why
Acetate is the major methane precursor in AD.
This is where:
biochar effect is modeled
Most important block.

RCSTR Kinetic Equation
Conceptually:

$$r = k(1 + \alpha B)C$$

where:

- k= base kinetic constant
- α= biochar enhancement factor
- B= biochar dose
- C= acetate concentration

Aspen uses simplified form:

$$r = k_{eff}C$$

where:

$$k_{eff} = k(1 + \alpha B)$$

Example
From Python:
Base k = 0.25
Alpha = 6.8
B = 0.05
Then:
k_eff = 0.335
This is entered in:
Power Law kinetics
inside RCSTR.

Your Current Setup
From your screenshot:
k = 0.35
n = 0
E = 0
Recommended improvement:
k = 0.35
n = 1
E = 0
because:

$$r = kC^n$$

and first-order reaction is more realistic.

Step 8 — MIXT Block

Purpose
Mix methane from:

- RSTOIC
- RCSTR

to create:
total biogas stream

Why
Final methane comes from both pathways.
So both must be combined.

Step 9 — SEP2 Block

Purpose
Separate final stream into:
ADGAS
Final methane-rich biogas
RESIDUE
Digestate + water + biochar

Why
To obtain final usable product stream.
This matches real digester output.

Final Outputs

ADGAS
Contains:

- CH_4
- CO_2
- trace gases

Used for:
methane yield validation

RESIDUE
Contains:

- digest
- water
- unreacted biochar

This proves:
biochar remains unreacted
Very strong thesis point.

Model Validation
Compare:
Aspen methane yield
with
experimental methane yield
Example:

Source	Methane Yield
Lab BMP	429
Aspen	417

Good agreement means:
model is valid

Final Scientific Conclusion
Biochar does not participate in chemical reactions.
It improves methane production only by increasing:
methanogenesis reaction kinetics
inside RCSTR.
This is the strongest scientific contribution of your work.